Emma Ugarcovici - Homework 4

April 3, 2026

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Loading required package: pacman
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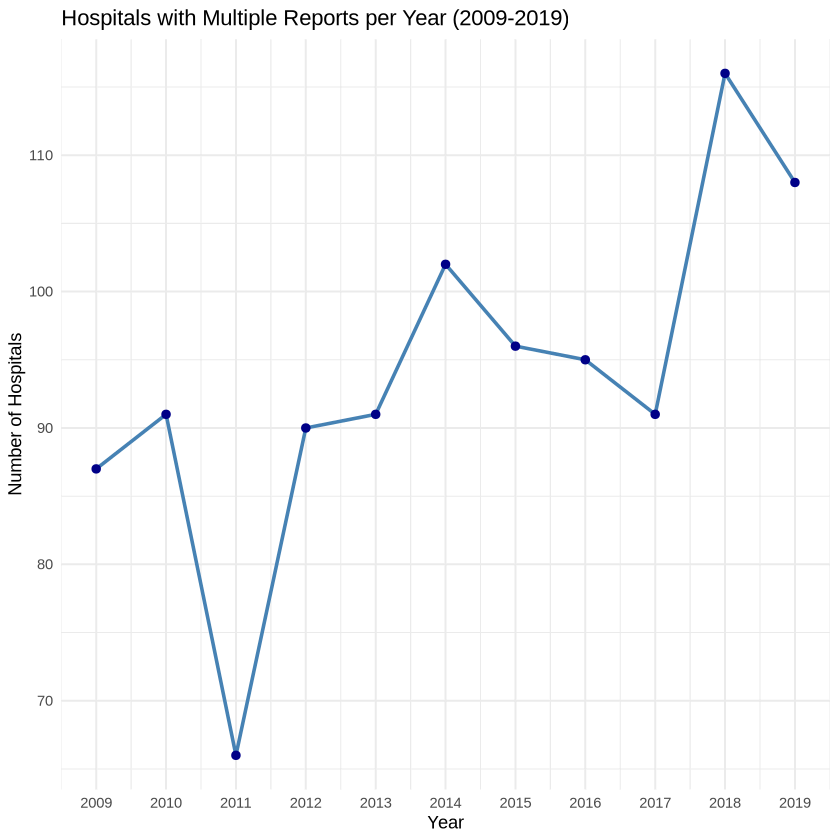
Question 1: From 2009-2019, how many hospitals filed more than one report in the same year?



Question 2: After removing/combining multiple reports, how many unique hospital IDs (Medicare provider numbers) exist in the data?

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Total unique Medicare provider numbers in cleaned data: 6897
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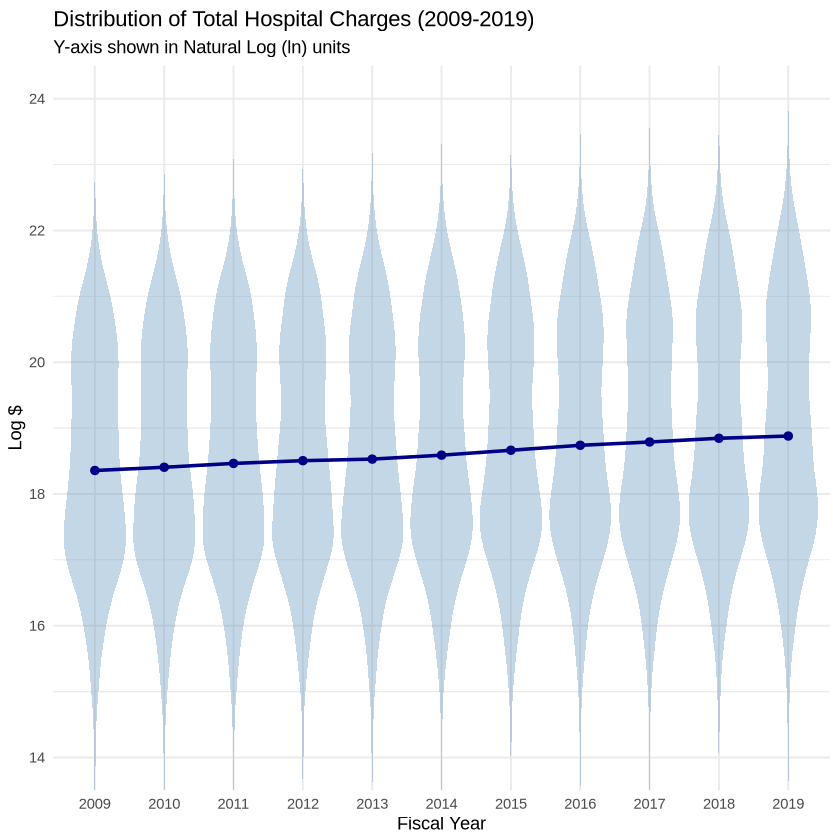
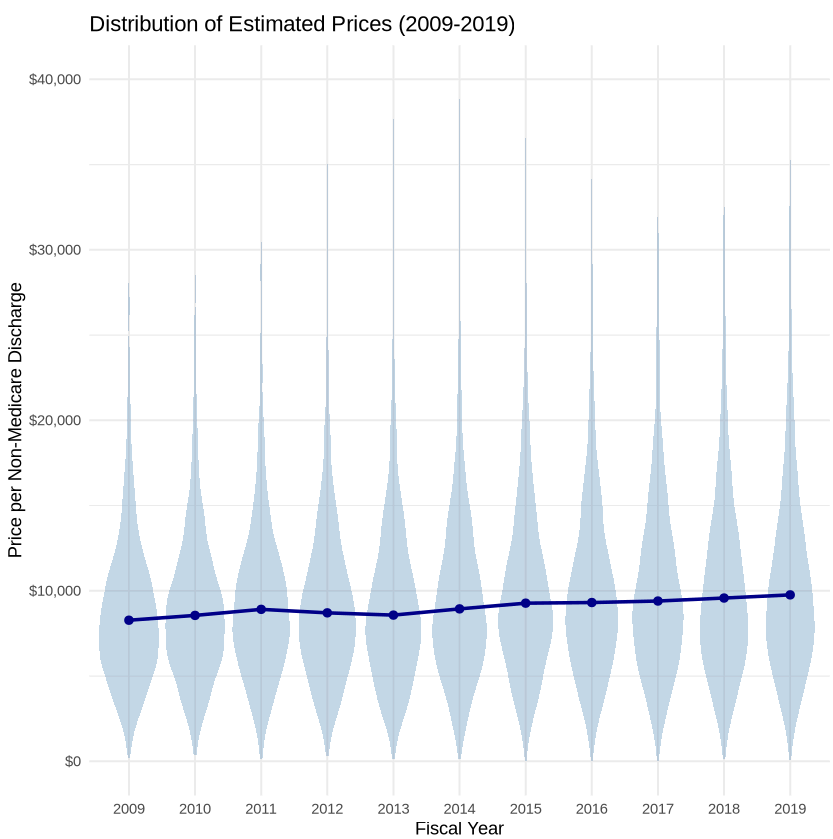
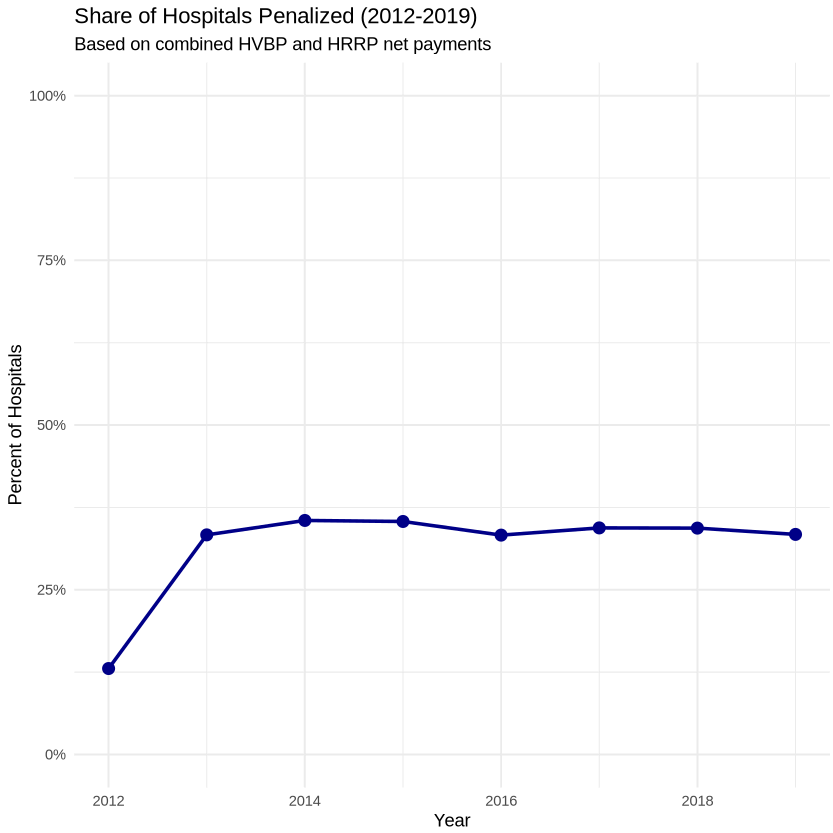
Question 3: From 2009-2019, what is the distribution of total charges (tot_charges in the data) in each year?



Question 4: From 2009-2019, what is the distribution of estimated prices in each year?



Question 5: What share of hospitals are penalized under the HRRP/VBP? (2012-2019)



Question 6: Provide a summary of OLS estimates of the effect of net penalties on price changes.

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NOTE: 5 observations removed because of NA values (RHS: 5).

NOTE: 5 observations removed because of NA values (RHS: 5).

NOTE: 16 observations removed because of NA values (RHS: 16).
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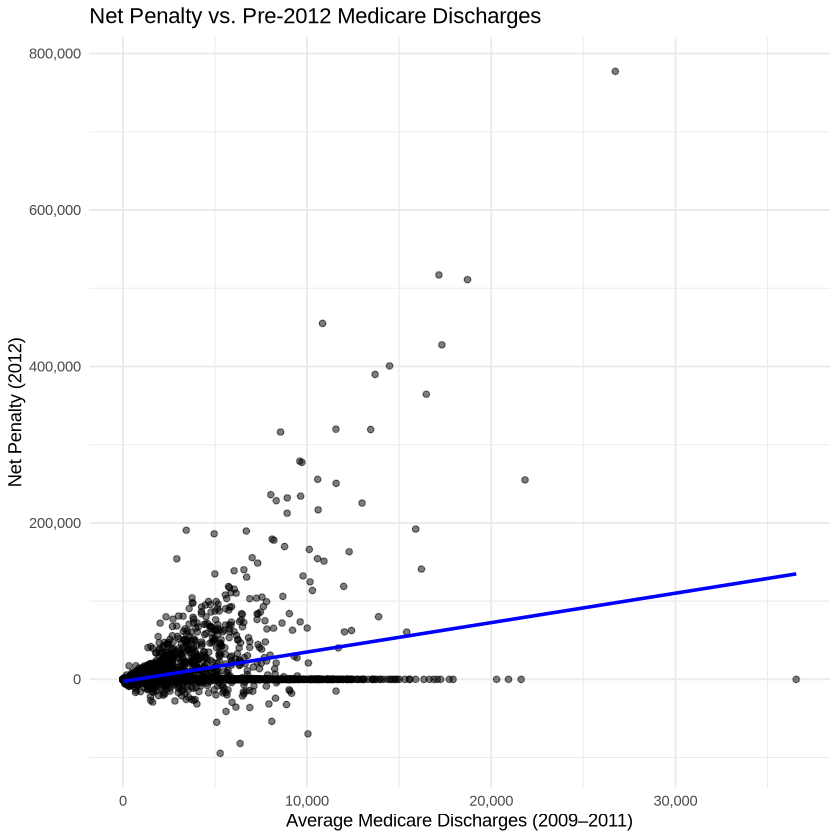
A etable_df: 11 × 4

		m1_ols	m2_ols	m3_ols
	<chr>	<chr>	<chr>	<chr>
1	Dependent Var.:	price_change	price_change	price_change
2				
3	Constant	84.98 (75.02)	66.95 (75.86)	-65.01 (93.50)
4	penalty_amt	-0.0045* (0.0019)	-0.0044* (0.0019)	-0.0039* (0.0019)
5	avg_beds		0.0751 (0.0476)	0.0668 (0.0478)
6	avg_mcaid_disch			0.0842* (0.0350)
7	_____	_____	_____	_____
8	S.E. type	IID	IID	IID
9	Observations	2,508	2,508	2,497
10	R2	0.00228	0.00327	0.00559
11	Adj. R2	0.00188	0.00247	0.00439

Question 7: Provide a scatterplot of net penalty against pre-2012 Medicare discharges.



Question 8: Provide a summary of the first stage and reduced-form results using pre-penalty Medicare discharges as an instrument for net penalties.

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NOTE: 5 observations removed because of NA values (LHS: 5).

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NOTE: 16 observations removed because of NA values (LHS: 5, RHS: 11).

NOTE: 11 observations removed because of NA values (RHS: 11).
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A etable_df: 12 × 4

		fs_1	fs_2	fs_3
	<chr>	<chr>	<chr>	<chr>
1		Baseline	+ Beds	+ Medicaid
2	Dependent Var.:	penalty_amt	penalty_amt	penalty_amt
3				
4	Constant	5,887.3*** (1,125.6)	5,885.1*** (1,126.3)	5,323.0*** (1,142.3)
5	avg_mcare_disch	-4.198*** (0.2350)	-4.200*** (0.2364)	-4.657*** (0.2716)
6	avg_beds		0.0327 (0.4820)	-0.0125 (0.4822)
7	avg_mcaid_disch			1.373*** (0.4016)
8	_____	_____	_____	_____
9	S.E. type	IID	IID	IID
10	Observations	2,508	2,508	2,497
11	R2	0.11292	0.11292	0.11692
12	Adj. R2	0.11257	0.11221	0.11586

A etable_df: 12 × 4

		rf_1	rf_2	rf_3
	<chr>	<chr>	<chr>	<chr>
1		Baseline	+ Beds	+ Medicaid
2	Dependent Var.:	price_change	price_change	price_change
3				
4	Constant	111.1 (111.7)	105.9 (111.7)	53.63 (113.4)
5	avg_mcare_disch	0.0046 (0.0233)	0.0005 (0.0235)	-0.0391 (0.0270)
6	avg_beds		0.0791. (0.0479)	0.0751 (0.0479)
7	avg_mcaid_disch			0.1204** (0.0399)
8	_____	_____	_____	_____
9	S.E. type	IID	IID	IID
10	Observations	2,513	2,513	2,502
11	R2	1.55e-5	0.00110	0.00473
12	Adj. R2	-0.00038	0.00031	0.00354

Question 9: Provide a summary of IV estimates of the effect of net penalties on price changes.

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NOTE: 11 observations removed because of NA values (RHS: 11).
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A etable_df: 11 × 4

		m1_iv	m2_iv	m3_iv
	<chr>	<chr>	<chr>	<chr>
1		Baseline	+ Beds	+ Medicaid
2	Dependent Var.:	price_change	price_change	price_change
3				
4	Constant	119.5 (89.62)	108.5 (89.89)	10.26 (99.90)
5	penalty_amt	-0.0008 (0.0056)	0.0002 (0.0056)	0.0088 (0.0058)
6	avg_beds		0.0790. (0.0479)	0.0752 (0.0483)
7	avg_mcaid_disch			0.1095** (0.0370)
8	_____	_____	_____	_____
9	S.E. type	IID	IID	IID
10	F-test	0.01951	1.3719	4.0069
11	Observations	2,508	2,508	2,497

Question 10: Briefly explain the “Local” ATE in the context of your estimates. How might a local effect differ from an overall ATE in this setting?

The IV estimate represents a Local ATE because it identifies the causal impact of net penalties on price changes specifically for the hospitals whose penalty status was directly influenced by their volume of Medicare discharges (the instrument). As discussed in class, hospitals with larger Medicare discharge volumes generate the most variation in the penalty variable within this framework. By using Medicare volume as an instrument, the LATE analysis focuses on hospitals whose penalties were more directly impacted by their exposure to this instrument rather than random factors.

There are notable differences between the OLS estimate of -0.0039 and the IV estimate of 0.0088. While the OLS suggests a tiny negative relationship, the IV suggests that for every dollar of net penalty, these highly-exposed hospitals increase private prices by approx 0.009. This subset likely differs from the average institution because they are more financially vulnerable and have smaller margins. If you only looked at the ATE, you might think that Medicare penalties don't impact prices. However, the LATE reveals that for the specific subset of hospitals actually targeted by the policy, the impact is significant. Focusing on the variation generated by Medicare discharges in the LATE filters out the noise of hospitals in the ATE that are indifferent to the penalty.